GEOLOGY | THE GEOLOGICAL SOCIETY OF AMERICA®

https://doi.org/10.1130/G47807.1

Manuscript received 28 April 2020
Revised manuscript received 13 October 2020
Manuscript accepted 19 October 2020

Published online 10 December 2020

© 2020 Geological Society of America. For permission to copy, contact editing@geosociety.org.

Inverted South China: A novel configuration for Rodinia and its breakup

Xianqing Jing[1], David A.D. Evans[2], Zhenyu Yang[1]*, Yabo Tong[3], Yingchao Xu[4] and Heng Wang[5]
[1]College of Resources, Environment and Tourism, Capital Normal University, Beijing 100048, China
[2]Department of Earth & Planetary Sciences, Yale University, New Haven, Connecticut 06520-8109, USA
[3]Institute of Geomechanics, Chinese Academy of Geological Sciences, Beijing 100081, China
[4]College of Geography and Tourism, Huanggang Normal University, Huanggang 438000, China
[5]Institute of Geology, China Earthquake Administration, Beijing 100029, China

ABSTRACT

Disentangling records of Rodinia fragmentation and true polar wander remains a challenge for understanding late Tonian plate tectonics. The ca. 760 Ma lower member of the Liántuó Formation, South China, yields a primary paleomagnetic remanence that passes both the fold and reversal tests. This new result and recently reported ca. 800 Ma data from elsewhere in South China suggest a new interpretation of its apparent polar wander path, whereby pre–770 Ma poles have inverted absolute polarity relative to traditional interpretations. Based on this inversion, and an interpretation of several oscillations of true polar wander documented by global data during 810–760 Ma, we propose a novel reconstruction for Rodinia and its breakup. Our reconstruction places the South China, India, and Kalahari cratons to the southwest of Laurentia, with connections that might have been established as early as ca. 1000 Ma. Our model also suggests that initial rifting of Rodinia occurred at ca. 800 Ma via fast northward motion of the India craton and South China.

INTRODUCTION

During the Tonian Period (1000–720 Ma), a time of important biological and climatic development (e.g., Kuznetsov et al., 2017), assembly and breakup of the supercontinent Rodinia were major features of Earth's plate-tectonic history (Li et al., 2013; Merdith et al., 2017). Coincident with Rodinia's tenure and breakup, several episodes of true polar wander (TPW), the uniform motion of solid Earth relative to its rotation axis, have been proposed (Maloof et al., 2006; Jing et al., 2020), though not universally adopted (Merdith et al., 2017). Oscillatory patterns of TPW might be expected if the global lithosphere had a significant effective elasticity (Creveling et al., 2012) or if the planet's inertial tensor were prone to rotational instability via symmetry of subduction zones (Evans, 1998; Robert et al., 2018).

Hypotheses of Rodinia reconstructions and putative Tonian TPW events rely largely on paleomagnetic data, but interpretations have hemispheric freedom due to unknown geomagnetic polarity and the possibility that the cratons' paleomagnetic apparent polar wander (APW) paths contain sizable gaps. Such is the case for South China's Tonian APW path, which recorded large degrees of motion between ca. 825 and 720 Ma (Jing et al., 2020). Recent compilation of the Tonian Laurentian APW path has documented oscillations through the same time interval (Eyster et al., 2020). In order to clarify the reconstruction of South China and Laurentia in Rodinia, and to explore hypotheses of Tonian TPW, we conducted a paleomagnetic study on the Tonian Lower Liántuó Formation in the western Three Gorges area (TGA) in central China, and we explored the implications of our results for our reconstruction, which considers both possible polarities of South China's Tonian data.

PALEOMAGNETIC RESULTS

We collected 128 samples (12 sites) from the Liántuó Formation lower member at Qīnglínkǒu village (30.801°N, 110.920°E), Húběi Province, central China (Fig. S1 in the Supplemental Material[1]). Details of the geological setting and paleomagnetism are provided in the Supplemental Material. Thermal demagnetization revealed a high-temperature component (HTC) between 550 °C and 670 °C (Fig. 1; Fig. S2). The HTC, together with previous results from the lower Liántuó Formation at the Tiánjiāyuánzí section (Jing et al., 2015), passes both the fold and reversal tests (Fig. 1). In addition, the pole (LTM1; Fig. 2; Fig. S3) derived from the HTC (20.0°N, 148.7°E; dp/dm = 8.7°/10.0° [for a 95% confidence ellipse, dp is the minor axis and dm is the major axis]; Table S1) is different from known Phanerozoic paleomagnetic poles of South China, but it is very close to the upper member Liántuó Formation pole (LTM2 in Fig. S3C; Jing et al., 2015). Together with the results of Jing et al. (2015), we suggest that the Liántuó Formation recorded at least seven geomagnetic polarity intervals during its deposition. Although clastic sedimentary rocks are commonly affected by remanent inclination shallowing (Kent and Tauxe, 2005), remanent anisotropy studies on the Liántuó Formation in the TGA demonstrate no significant shallowing (Jing et al., 2016). Anisotropy of magnetic susceptibility data also suggest no significant later structural effects on our section (Fig. S4). Three tuff bed zircon U-Pb ages from the base of the Liántuó Formation are 763 ± 10 Ma, 770 ± 6 Ma, and 767 ± 3 Ma (Lan et al., 2015; Jing et al., 2018). Constrained further by a 748 ± 12 Ma tuff bed zircon U-Pb age from the top of the Liántuó Formation lower member, we assigned an age of 760 ± 10 Ma to our new LTM1 pole.

*E-mail: zhenyu.yang@cnu.edu.cn

[1]Supplemental Material. Data presented in the paper, detailed geological setting of the sampling section, laboratory methods, paleomagnetic results, and data analysis. Please visit https://doi.org/10.1130/GEOL.S.13262831 to access the supplemental material, and contact editing@geosociety.org with any questions.

CITATION: Jing, X., et al., 2021, Inverted South China: A novel configuration for Rodinia and its breakup: Geology, v. 49, p. 463–467, https://doi.org/10.1130/G47807.1

Downloaded from http://pubs.geoscienceworld.org/gsa/geology/article-pdf/49/4/463/5252932/463.pdf
by Curtin University Library user

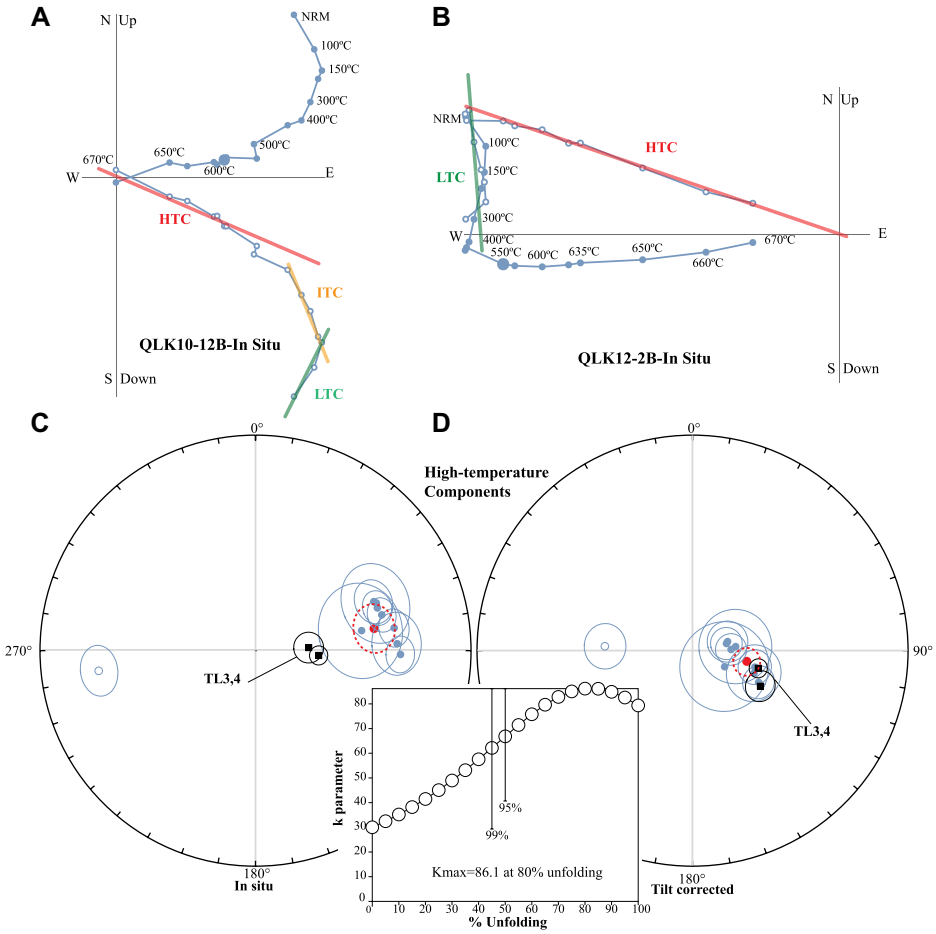


Figure 1. (A,B) Orthogonal vector projection of thermal demagnetization for representative samples from this study (from the Tonian Lower Liántuó Formation in the western Three Gorges area, central China), *in situ* coordinates. Red, yellow, and green lines are fitted high-, intermediate-, and low-temperature components (HTC, ITC, and LTC), respectively. Filled/open blue circles correspond to points in the horizontal/vertical planes, respectively. NRM—natural remanent magnetization. (C,D) Site-mean directions of HTC *in situ* and in stratigraphic coordinates. Filled/open symbols represent directions in the lower/upper hemisphere, respectively. Blue symbols are results of this study; black squares are data reported by Jing et al. (2015). Red symbols are Fisher means of site data, with the dashed red circle as the 95% confidence level. Inset shows stepwise unfolding with Fisher's precision parameter (*k*) plotted against the level of unfolding.

INVERTED SOUTH CHINA AND OSCILLATORY TPW

Traditionally, all Tonian poles from South China have been assigned a polarity such that the craton would have remained in the Northern Hemisphere from ca. 820 Ma (Li et al., 2004; Niu et al., 2016; Jing et al., 2020) to culminate with the ca. 720 Ma mid-paleolatitude LTM2 pole (Jing et al., 2015). The trajectory of these poles appears to circumscribe a large loop (Fig. S3C), but two recently reported younger poles (YB-B and CJH3) (Niu et al., 2016; Jing et al., 2020) are separated from LTM2 by ~112° (Fig. S3C). Although a precise age for CJH3 is lacking, it should be between 799 ± 8 and 785 ± 12 Ma, as measured by sensitive high-resolution ion microprobe (SHRIMP) U-Pb ages on zircons from tuffs below and above the CJH3 strata (Jing et al., 2020). The LTM1 pole lies ~100° away from poles YB-B (806 ± 8 Ma) and CJH3 (Fig. S3C). Even considering geochronological uncertainty, these data still indicate a plate speed of ≥20 cm/yr for South China across the northern polar regions if the traditional polarity choice is retained for the older poles.

Alternatively, we can invert the polarity of all pre–770 Ma paleopoles from South China. In so doing, we obtain a unidirectional Tonian APW path for South China, and the angular distance between the CJH3 and LTM1 poles is reduced to ~80° (Fig. 2A; Fig. S3C). This still requires South China to move with a velocity of ≥14 cm/yr, which is fast for a tectonic plate. Recently, Eyster et al. (2020) summarized 780–720 Ma poles from Laurentia (Fig. 2B; Table S2). New age constraints on these poles also suggested fast rotational movement of Laurentia during 775–766 Ma, with APW rates of ≈30 cm/yr. Considering Laurentia as the core of the Rodinia supercontinent, such fast rotation is uncomfortably high if solely interpreted as plate movement. Therefore, we suggest that the rapid motion documented from both Laurentia and South China during ca. 780–760 Ma may also have been dominated by TPW.

Intriguingly, this proposed TPW migration is coaxial with putative TPW events from the closely preceding interval, 810–790 Ma, when Svalbard was restored adjacent to northern Greenland (Fig. 2B). We thus propose three phases of oscillatory TPW. Phase I occurred during 810–800 Ma, recorded by Svalbard (Laurentia) and South China (Figs. 2A and 2B; Table 1; Maloof et al., 2006; Jing et al., 2020). Phase II, with an opposite sense of motion, occurred during 800–790 Ma (Maloof et al., 2006); in the APW path, this phase is currently only reflected by Svalbard poles, as we hypothesize that South China was independently moving in the opposite sense during that time (see below). Phase III is the TPW event we suggest here during 780–760 Ma, recorded by both Laurentia and South China (Figs. 2A and 2B; Table 1).

One of the predictions of the TPW is that it can induce sea-level change (Mound et al., 1999). Sea-level variation matches the TPW events during 810–790 Ma for east Svalbard (Maloof et al., 2006). Stratigraphic and geochronological studies (Chen et al., 2016) suggest that South China experienced transgression during 810–800 Ma and a gradual regression during 800–760 Ma, which matches our proposed multiple TPW events. During phase I, South China moved from the polar zone to middle latitudes, which induced transgression. Phase II was in the opposite sense, but South China's individual plate motion maintained moderate latitudes with little net effect on freeboard. During phase III, South China traveled across the equator to higher latitudes, which would have resulted in regression.

NOVEL SOUTH CHINA AND RODINIA RECONSTRUCTION

If oscillatory TPW occurred during 825–760 Ma, the paleomagnetic results during that time span from each continent should distribute along a great circle, orthogonal to the planet's stable axis of minimum inertial moment (I_{min}; Evans, 2003; Mitchell et al., 2012). Assuming insignificant rates of relative plate motion, those circles can be superimposed to generate reconstructions constrained in relative paleolongitude as well as paleolatitude (Raub et al., 2007). In an ideal supercontinental cycle, I_{min} should approximate the center of a fragmenting supercontinent (Evans, 2003). The three phases during ca. 810–760 Ma TPW proposed above are proposed to have oscillated about the I_{min} associated with Rodinia's breakup. Individual cratons were reconstructed in paleolongitude relative to this I_{min}, as follows.

Downloaded from http://pubs.geoscienceworld.org/gsa/geology/article-pdf/49/4/463/5252932/463.pdf
by Curtin University Library user

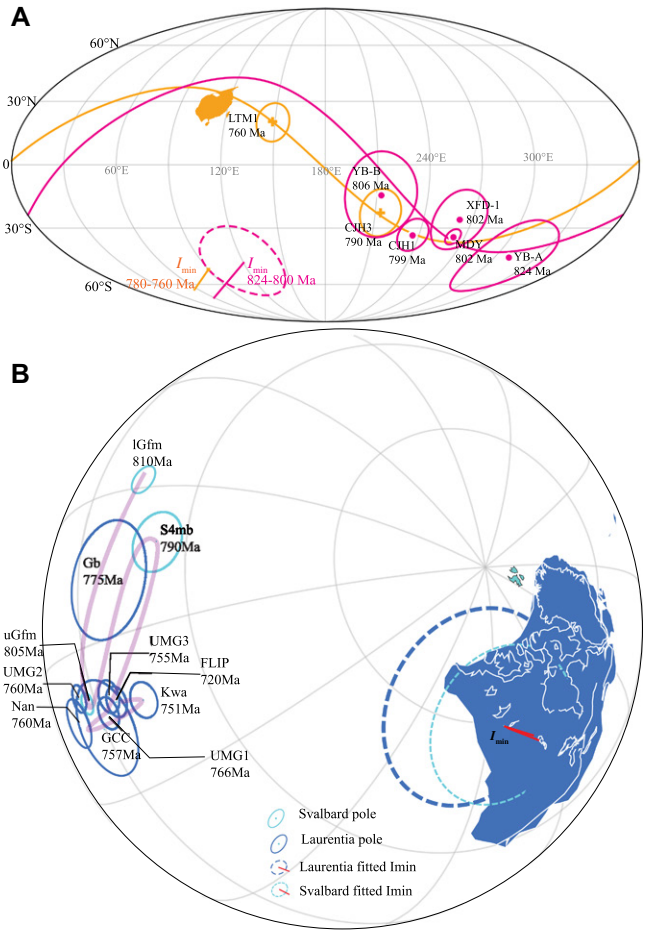


Figure 2. (A) Inverted South China poles (orange circles with crosses and magenta circles with dots) and two great circles fitted to 825–800 Ma (magenta) and 780–760 Ma (orange) poles. Magenta/orange dashed circles with central lines are fitted minimum inertial moment axes (I_{min}) with 95% confidence ellipses. (B) 810–720 Ma paleomagnetic poles for Laurentia (dark blue) and Svalbard (light blue). Svalbard is restored to north Greenland (see text). Purple curve is the combined apparent polar wander path of Laurentia during 810–720 Ma. See the Supplemental Material (see footnote 1) for paleomagnetic poles.

Svalbard is restored to the north of Greenland by overlapping the fitted I_{min} of Laurentia (780–760 Ma) and Svalbard (810–790 Ma), and by positioning the 780 Ma Gunbarrel (Gb) pole close to the Svanbergfjellet member 4 (S4mb) pole (Fig. 2B; Euler parameters are provided in Table S3). Svalbard's location north of Greenland differs from that proposed by Maloof et al. (2006) but is similar to the placement illustrated by Gee and Teben'kov (2004). Our first-order conclusions are sensitive to the azimuthal orientation of Svalbard but not as much to its reconstructed location along the Greenland margin. The reconstruction of Laurentia atop I_{min} suggests its position near the core of Rodinia, as traditionally acknowledged (Hoffman, 1991). During 810–760 Ma, Laurentia had no significant motion relative to the I_{min} (Fig. 2B). The constancy of Laurentia's position atop a Rodinia-central I_{min} axis suggests that Laurentia's APW path might represent an essentially "pure" signal of oscillatory TPW through this interval, with an insignificant component of individual plate motion.

As noted above, South China's 820–760 Ma poles also lie largely on great circles, regardless of geomagnetic polarity interpretation, and we interpret its APW path as a combined record of oscillating TPW and plate motion (Fig. 2A; Jing et al., 2020). For the hypothesized ca. 810–800 Ma TPW event, overlapping the best-fitting I_{min} axes and great circles from South China and Svalbard reconstructs those two blocks far apart from each other; in our proposed geomagnetic polarity choice, South China would reconstruct to high southerly paleolatitudes (Figs. 3A and 3B). This reconstruction leaves a gap between South China and Laurentia (Fig. 3B), but the gap can be filled by the India and Kalahari cratons, based on several lines of evidence. First, several studies have suggested that South China was a part of the Greater India continent beginning in the Mesoproterozoic (Zhang et al., 2020). Second, coeval ca. 1110 Ma large igneous provinces (LIPs) in the Kalahari and India cratons, and the paleomagnetic studies on those LIPs, support a direct connection between these two cratons and place them far southwest of Laurentia, as part of a hypothesized continent named "Umkondia" (Choudhary et al., 2019). Swanson-Hysell et al. (2019) recently applied Bayesian fit to the Keweenawan track of the Laurentian APW path and suggested that Laurentia underwent fast plate motion (>20 cm/yr) during 1110–1090 Ma, converging toward Umkondia, and slowing dramatically upon ca. 1090 Ma collision. In our reconstruction, we restore the India craton by overlapping the 760 Ma Malani Igneous Suite (MIS) pole with the LTM1 pole, and we place South China to the current west of the India craton (Figs. 3A and 3B). The Kalahari craton is restored by juxtaposing its Namaqua belt with the Grenville belt, and its Umkondo LIP close to the coeval Indian mafic dike swarm.

The Australia craton is restored to western Laurentia by inserting the 830–800 Ma Browne Formation (Brf) pole between the ca. 820 Ma YB-A and 802 Ma XFD-1 and MDY poles from South China, and also overlapping the ca. 770 Ma Johnny's Creek Member (JCM) pole with the ca. 775 Ma Gunbarrel (Gb) pole of Laurentia (Fig. 3A). However, this leaves a gap between Australia and Laurentia (Fig. 3B). Instead of placing South China between Australia and Laurentia, Wen et al. (2018) suggested that Tarim and its ancestral neighbors may have formed the so-called "missing link," which we adopt in our model (Fig. 3B).

As noted here, the TPW event comparison can constrain the relative positions of continents, even if they were separate plates (in comparatively slow relative motion). In the traditional geomagnetic polarity interpretation of ca. 760 Ma poles from Laurentia and the India craton, the latter would have lain at moderate northerly paleolatitudes at that time (e.g. Meert et al., 2017). Following that convention, we place South China adjacent to India according to our new pole LTM1. Then, the superimposition of the ca. 780–760 Ma APW arcs (and corresponding I_{min}) of South China and Laurentia—interpreted as phase III of TPW—places South China and India far northwest of Laurentia

TABLE 1. DETAILS OF PROPOSED TRUE POLAR WANDER (TPW) OSCILLATION EVENTS DURING 810–760 MA

Phase	Duration	Amplitude (angular distance: Laurentia/South China)	Rodinia's sense of rotation via TPW	Area of record	Poles used for calculation (listed in Table S2*)	Latitudinal change of South China (reference point: 30.8°N, 110.9°E)
I	Ca. 810–800 Ma	≥45°/≥56°	Clockwise	East Svalbard (Laurentia); South China	lGfm, uGfm; XFD-1, YB-A, MDY, CJH1	69.1°S to 44.1°S
II	Ca. 800–790 Ma	≥36°/N.A.†	Counterclockwise	East Svalbard (Laurentia)	uGfm, S4mb	44.1°S to 22.0°S†
III	Ca. 780–760 Ma	≥25°/≥76°*†	Clockwise	Laurentia; South China	Gb, UGM1; CJH3, YB-B, LTM1	22.0°S to 54.4°N†

*See text footnote 1.
†During proposed TPW phases II and III, South China's aggregate motion includes the combined effects of TPW and plate motion following Rodinia breakup.

Downloaded from http://pubs.geoscienceworld.org/gsa/geology/article-pdf/49/4/463/5252932/463.pdf
by Curtin University Library user

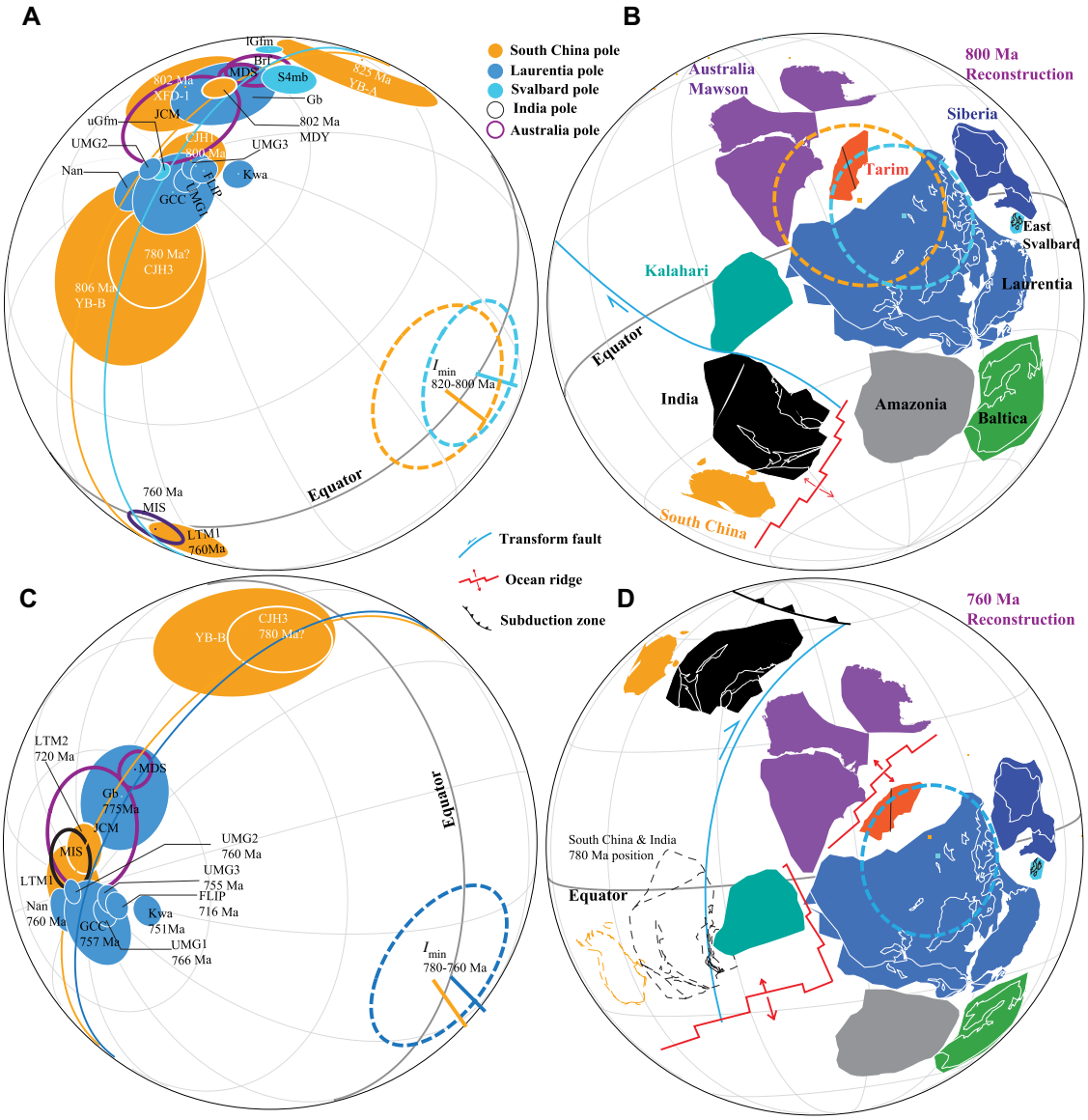


Figure 3. (A) Apparent polar wander (APW) path comparison of South China (orange), and the Laurentia and Svalbard (blue), Australia (purple), and India (black) cratons, demonstrating that after ca. 800 Ma, APW paths of South China and Rodinia move in opposite directions. See the Supplemental Material (see footnote 1) for paleomagnetic poles. (B) Proposed reconstruction of Rodinia at 800 Ma. (C) APW comparison of South China, Laurentia and Svalbard, Australia, and India cratons during 780–720 Ma. (D) Proposed reconstruction of fragmenting Rodinia at 760 Ma. Dash-outlined South China and the India craton are ca. 780 Ma reconstruction positions. Positions of Siberia, Baltica, and Amazonia are after Evans et al. (2016), Cawood and Pisarevsky (2017), and Gong et al. (2018). Blue/orange dashed circles with central lines are fitted inertial moment axes, I_{min}, with 95% confidence ellipses.

(Figs. 3C and 3D). Therefore, South China, and by inference India, should have broken away from Rodinia before 760 Ma, having drifted rapidly across the paleo-equator after 800 Ma. This motion is in the opposite sense to phase II of inferred TPW (800–790 Ma), and so the overall APW for South China is subdued by the two counteracting components of motion during that phase. Large-scale numerical simulations and tectonic reconstructions suggest that plates can move fast during initial breakup and can retain such speeds for tens of millions of years (Ulvrova et al., 2019). This may explain the rapidity of South China's northward motion following earliest Rodinia breakup (Fig. 3). The numerical simulations also suggest distal subduction forces as a means to generate rifting (Ulvrova et al., 2019). In our model, we speculate that a Northern Hemisphere subduction zone initiated at ca. 810–800 Ma, triggering the rifting and rapid northward motion of South China and India during earliest Rodinia breakup (Figs. 3B and 3D).

Our new reconstruction suggests that the initial rifting of Rodinia occurred in the Southern Hemisphere via largely transcurrent motion of the India craton and South China (Figs. 3B and 3D). This tectonic transition is not only reflected by the divergence of the APW paths of South China and Rodinia at ca. 800 Ma (Fig. 3A), but it is also recorded by a coeval transition in $\delta^{18}O$ and $\varepsilon_{Hf(t)}$ records from South China and the India craton, interpreted as a rift signature (Wang et al., 2017). In addition, ca. 800–790 Ma A-type magmatism reported from the Eastern Ghats Belt of the India craton (Ranjan et al., 2018) and the northern and western margins of the Kalahari craton (Jacobs et al., 2008; Rioux et al., 2010) may signify Rodinia fragmentation in its southwestern sector.

CONCLUSION

Under traditional interpretations of hemispheric assignment to paleomagnetic poles, our new paleomagnetic result would require an extremely fast plate motion of South China during ca. 780–760 Ma. However, inversion of South China's pre–770 Ma APW path can mitigate this problem. Based on the inverted APW path for South China, and in comparison to global paleomagnetic and tectonic records of the late Tonian, we produced an integrated model of initial Rodinia fragmentation in the context of oscillatory TPW. Our new reconstruction model for Rodinia locates South China and the India, and Kalahari cratons to the southwest of Laurentia. Upon Rodinia's initial breakup at ca. 800 Ma, South China and India drifted transcurrently relative to Laurentia, and moved rapidly to the Northern Hemisphere, setting the stage for subsequent amalgamation of East Gondwanaland.

ACKNOWLEDGMENTS

We thank Sheree Armistead, Roger Fu, Ben Weiss, and an anonymous reviewer, and editor Dennis Brown, for their insightful comments on the manuscript. This work was supported by the National

Downloaded from http://pubs.geoscienceworld.org/gsa/geology/article-pdf/49/4/463/5252932/463.pdf
by Curtin University Library user

Natural Science Foundation of China (grant 41230208). Xianqing Jing was supported by the China Scholarship Council (no. 201808110036).

REFERENCES CITED

Cawood, P.A., and Pisarevsky, S.A., 2017, Laurentia-Baltica-Amazonia relations during Rodinia assembly: Precambrian Research, v. 292, p. 386–397, https://doi.org/10.1016/J.PRECAMRES.2017.01.031.

Chen, J., Dai, C., Peng, C., Lu, D., Wang, X., Wang, M., Bao, L., Zhang, D., and Luo, S., 2016, Study on stratigraphical division and correlation of the Neoproterozoic "Xiajiang Group" in Hunan, Guizhou and Guangxi Provinces: Discussion on the reboot of Xiajiang system: Geological Review (Dizhi Lunping), v. 62, p. 1093–1114 [in Chinese with English abstract], https://doi.org/10.16509/j.georeview.2016.05.001.

Choudhary, B.R., Ernst, R.E., Xu, Y.-G., Evans, D.A.D., de Kock, M., Meert, J.G., Ruiz, A., and Lima, G.A., 2019, Geochemical characterization of a reconstructed 1110 Ma large igneous province: Precambrian Research, v. 332, p. 105382, https://doi.org/10.1016/j.precamres.2019.105382.

Creveling, J.R., Mitrovica, J.X., Chan, N.-H., Latychev, K., and Matsuyama, I., 2012, Mechanisms for oscillatory true polar wander: Nature, v. 491, p. 244–248, https://doi.org/10.1038/nature11571.

Evans, D.A.D., 1998, True polar wander: A supercontinental legacy: Earth and Planetary Science Letters, v. 157, p. 1–8, https://doi.org/10.1016/S0012-821X(98)00031-4.

Evans, D.A.D., 2003, True polar wander and supercontinents: Tectonophysics, v. 362, p. 303–320, https://doi.org/10.1016/S0040-1951(02)000642-X.

Evans, D.A.D., Veselovsky, R.V., Petrov, P.Y., Shatsillo, A.V., and Pavlov, V.E., 2016, Paleomagnetism of Mesoproterozoic margins of the Anabar Shield: A hypothesized billion-year partnership of Siberia and northern Laurentia: Precambrian Research, v. 281, p. 639–655, https://doi.org/10.1016/j.precamres.2016.06.017.

Eyster, A., Weiss, B.P., Karlstrom, K., and Macdonald, F.A., 2020, Paleomagnetism of the Chuar Group and evaluation of the late Tonian Laurentian apparent polar wander path with implications for the makeup and breakup of Rodinia: Geological Society of America Bulletin, v. 132, p. 710–738, https://doi.org/10.1130/B32012.1.

Gee, D.G., and Teben'kov, A.M., 2004, Svalbard: A fragment of the Laurentian margin, *in* Gee, D.G., and Pease, V., eds., The Neoproterozoic Timanide Orogen of Eastern Baltica: Geological Society [London] Memoir 30, p. 191–206, https://doi.org/10.1144/GSL.MEM.2004.030.01.16.

Gong, Z., Evans, D.A.D., Elming, S.-Å., Söderlund, U., and Salminen, J.M., 2018, Paleomagnetism, magnetic anisotropy and U-Pb baddeleyite geochronology of the early Neoproterozoic Blekinge-Dalarna dolerite dykes, Sweden: Precambrian Research, v. 317, p. 14–32, https://doi.org/10.1016/j.precamres.2018.08.019.

Hoffman, P.F., 1991, Did the breakout of Laurentia turn Gondwanaland inside-out?: Science, v. 252, p. 1409–1412, https://doi.org/10.1126/science.252.5011.1409.

Jacobs, J., Pisarevsky, S., Thomas, R.J., and Becker, T., 2008, The Kalahari craton during the assembly and dispersal of Rodinia: Precambrian Research, v. 160, p. 142–158, https://doi.org/10.1016/j.precamres.2007.04.022.

Jing, X., Yang, Z., Tong, Y., and Han, Z., 2015, A revised paleomagnetic pole from the mid-Neoproterozoic Liantuo Formation in the Yangtze block and its paleogeographic implications: Precambrian Research, v. 268, p. 194–211, https://doi.org/10.1016/j.precamres.2015.07.007.

Jing, X., Yang, Z., Tong, Y., Wang, H., Han, Z., and Xu, Y., 2016, Inclination shallowing study of the early Neoproterozoic Liantuo Formation in south China and its paleogeographic implications: Geological Bulletin of China, v. 35, p. 1797–1806 [in Chinese with English abstract].

Jing, X., Yang, Z., Tong, Y., Wang, H., and Xu, Y., 2018, A SHRIMP U-Pb zircon geochronology of a tuff bed from the bottom of Liantuo Formation in the Three Gorges area and its geological implications: Journal of Jilin University, v. 48, p. 165–180, https://doi.org/10.13278/j.cnki.jjuese.20160232.

Jing, X., Yang, Z., Evans, D.A.D., Tong, Y., Xu, Y., and Wang, H., 2020, A pan-latitudinal Rodinia in the Tonian true polar wander frame: Earth and Planetary Science Letters, v. 530, p. 115880, https://doi.org/10.1016/j.epsl.2019.115880.

Kent, D.V., and Tauxe, L., 2005, Corrected Late Triassic latitudes for continents adjacent to the North Atlantic: Science, v. 307, p. 240–244, https://doi.org/10.1126/science.1105826.

Kuznetsov, A.B., Bekker, A., Ovchinnikova, G.V., Gorokhov, I.M., and Vasilyeva, I.M., 2017, Unradiogenic strontium and moderate-amplitude carbon isotope variations in early Tonian seawater after the assembly of Rodinia and before the Bitter Springs excursion: Precambrian Research, v. 298, p. 157–173, https://doi.org/10.1016/j.precamres.2017.06.011.

Lan, Z., Li, X.H., Zhu, M., Zhang, Q., and Li, Q.L., 2015, Revisiting the Liantuo Formation in Yangtze block, south China: SIMS U-Pb zircon age constraints and regional and global significance: Precambrian Research, v. 263, p. 123–141, https://doi.org/10.1016/j.precamres.2015.03.012.

Li, Z.-X., Evans, D.A.D., and Zhang, S., 2004, A 90° spin on Rodinia: Possible causal links between the Neoproterozoic supercontinent, superplume, true polar wander and low-latitude glaciation: Earth and Planetary Science Letters, v. 220, p. 409–421, https://doi.org/10.1016/S0012-821X(04)00064-0.

Li, Z.-X., Evans, D.A.D., and Halverson, G.P., 2013, Neoproterozoic glaciations in a revised global palaeogeography from the breakup of Rodinia to the assembly of Gondwanaland: Sedimentary Geology, v. 294, p. 219–232, https://doi.org/10.1016/j.sedgeo.2013.05.016.

Maloof, A.C., Halverson, G.P., Kirschvink, J.L., Schrag, D.P., Weiss, B.P., and Hoffman, P.F., 2006, Combined paleomagnetic, isotopic, and stratigraphic evidence for true polar wander from the Neoproterozoic Akademikerbreen Group, Svalbard, Norway: Geological Society of America Bulletin, v. 118, p. 1099–1124, https://doi.org/10.1130/B25892.1.

Meert, J.G., Pandit, M.K., Pivarunas, A., Katusin, K., and Sinha, A.K., 2017, India and Antarctica in the Precambrian: A brief analysis, *in* Pant, N.C., and Dasgupta, S., eds., Crustal Evolution of India and Antarctica: The Supercontinent Connection: Geological Society [London] Special Publication 457, p. 339–351, https://doi.org/10.1144/SP457.13.

Merdith, A.S., Collins, A.S., Williams, S.E., Pisarevsky, S., Foden, J.D., Archibald, D.B., Blades, M.L., Alessio, B.L., Armistead, S., Plavsa, D., Clark, C., and Müller, R.D., 2017, A full-plate global reconstruction of the Neoproterozoic: Gondwana Research, v. 50, p. 84–134, https://doi.org/10.1016/j.gr.2017.04.001.

Mitchell, R.N., Kilian, T.M., and Evans, D.A.D., 2012, Supercontinent cycles and the calculation of absolute palaeolongitude in deep time: Nature, v. 482, p. 208–211, https://doi.org/10.1038/nature10800.

Mound, J.E., Mitrovica, J.X., Evans, D.A.D., and Kirschvink, J.L., 1999, A sea-level test for inertial interchange true polar wander events: Geophysical Journal International, v. 136, https://doi.org/10.1046/j.1365-246X.1999.00791.x.

Niu, J., Li, Z.-X., and Zhu, W., 2016, Palaeomagnetism and geochronology of mid-Neoproterozoic Yanbian dykes, south China: Implications for a c. 820–800 Ma true polar wander event and the reconstruction of Rodinia, *in* Li, Z.-X., Evans, D.A.D., and Murphy, J.B., eds., Supercontinent Cycles Through Earth History: Geological Society [London] Special Publication 424, p. 191–211, https://doi.org/10.1144/SP424.11.

Ranjan, S., Upadhyay, D., Abhinay, K., Pruseth, K.L., and Nanda, J.K., 2018, Zircon geochronology of deformed alkaline rocks along the Eastern Ghats Belt margin: India–Antarctica connection and the Enderbia continent: Precambrian Research, v. 310, p. 407–424, https://doi.org/10.1016/J.PRECAMRES.2018.04.005.

Raub, T.D., Kirschvink, J.L., and Evans, D.A.D., 2007, True polar wander: Linking deep and shallow geodynamics to hydro- and biospheric hypotheses, *in* Elias, S., ed., Reference Module in Earth Systems and Environmental Sciences: Treatise on Geophysics, v. 5, p. 511–530, https://doi.org/10.1016/B978-0-444-53802-4.00108-1.

Rioux, M., Bowring, S., Dudás, F., and Hanson, R., 2010, Characterizing the U-Pb systematics of baddeleyite through chemical abrasion: Application of multi-step digestion methods to baddeleyite geochronology: Contributions to Mineralogy and Petrology, v. 160, p. 777–801, https://doi.org/10.1007/s00410-010-0507-1.

Robert, B., Greff-Lefftz, M., and Besse, J., 2018, True polar wander: A key indicator for plate configuration and mantle convection during the late Neoproterozoic: Geochemistry Geophysics Geosystems, v. 19, p. 3478–3495, https://doi.org/10.1029/2018GC007490.

Swanson-Hysell, N.L., Ramezani, J., Fairchild, L.M., and Rose, I.R., 2019, Failed rifting and fast drifting: Midcontinent rift development, Laurentia's rapid motion and the driver of Grenvillian orogenesis: Geological Society of America Bulletin, v. 131, no. 5–6, p. 913–940, https://doi.org/10.1130/B31944.1.

Ulvrova, M.M., Brune, S., and Williams, S., 2019, Breakup without borders: How continents speed up and slow down during rifting: Geophysical Research Letters, v. 46, p. 1338–1347, https://doi.org/10.1029/2018GL080387.

Wang, W., Cawood, P.A., Zhou, M.-F., Pandit, M.K., Xia, X.-P., and Zhao, J.-H., 2017, Low-$\delta^{18}O$ rhyolites from the Malani igneous suite: A positive test for south China and NW India linkage in Rodinia: Geophysical Research Letters, v. 44, p. 10298–10305, https://doi.org/10.1002/2017GL074717.

Wen, B., Evans, D.A.D., Wang, C., Li, Y.-X., and Jing, X., 2018, A positive test for the Greater Tarim block at the heart of Rodinia: Mega-dextral suturing of supercontinent assembly: Geology, v. 46, p. 687–690, https://doi.org/10.1130/G40254.1.

Zhang, L., Cawood, P.A., Wang, Y., Cui, X., Zhang, Y., Qian, X., and Zhang, F., 2020, Provenance record of late Mesoproterozoic to early Neoproterozoic units, west Hainan, south China, and implications for Rodinia reconstruction: Tectonics, v. 39, p. e2020TC006071, https://doi.org/10.1029/2020tc006071.

Printed in USA

Downloaded from http://pubs.geoscienceworld.org/gsa/geology/article-pdf/49/4/463/5252932/463.pdf
by Curtin University Library user